

# Results

01 • MANOVA

groups compared on several outcomes at once

**Table 1.** Multivariate tests

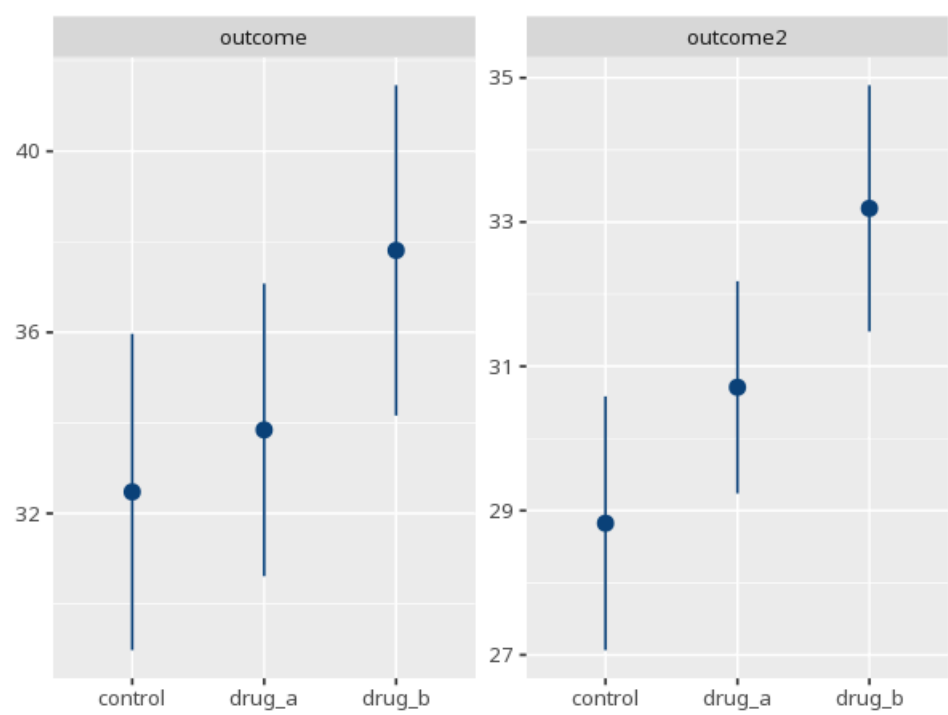
| Effect | Pillai / Wilks | approx F | df1 | df2 | p    |
|--------|----------------|----------|-----|-----|------|
| group  | 0.29           | 4.74     | 4   | 114 | .001 |

assumption check: homogeneity of covariance matrices (Box's M). (Box's M  $\chi^2=11.44$ ,  $df=6$ ,  $p=.076$ )

**Table 2.** Follow-up univariate ANOVAs (per DV)

| DV       | F    | df1 | df2 | p    | partial $\eta^2$ [95% CI] |
|----------|------|-----|-----|------|---------------------------|
| outcome  | 2.81 | 2   | 57  | .069 | 0.09 [0.00, 1.00]         |
| outcome2 | 7.71 | 2   | 57  | .001 | 0.21 [0.06, 1.00]         |

**Figure.** Group means per outcome



**How to read this test**

Tests whether groups differ on a set of outcomes jointly. Read the multivariate p (Pillai's trace is robust) first; if significant, the per-DV follow-up ANOVAs show which individual outcomes drive it. Because you run one ANOVA per outcome, correct those follow-up p-values for multiple comparisons (e.g. Bonferroni: divide alpha by the number of outcomes) before calling each significant. Your significance threshold ( $\alpha$ ) is 0.05.

**APA template:** A MANOVA gave Pillai's  $V=.29$ ,  $F(4,114)=4.74$ ,  $p = .001$ .

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